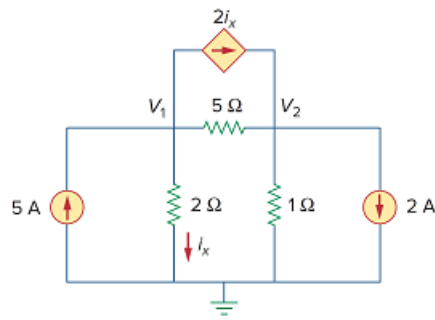


Problem

Solve for V_1 and V_2 in the circuit of Fig. 3.119 using *PSpice* or *MultiSim*.

Figure 3.119:



Step-by-step solution

Step 1 of 8

Consider the following circuit diagram:

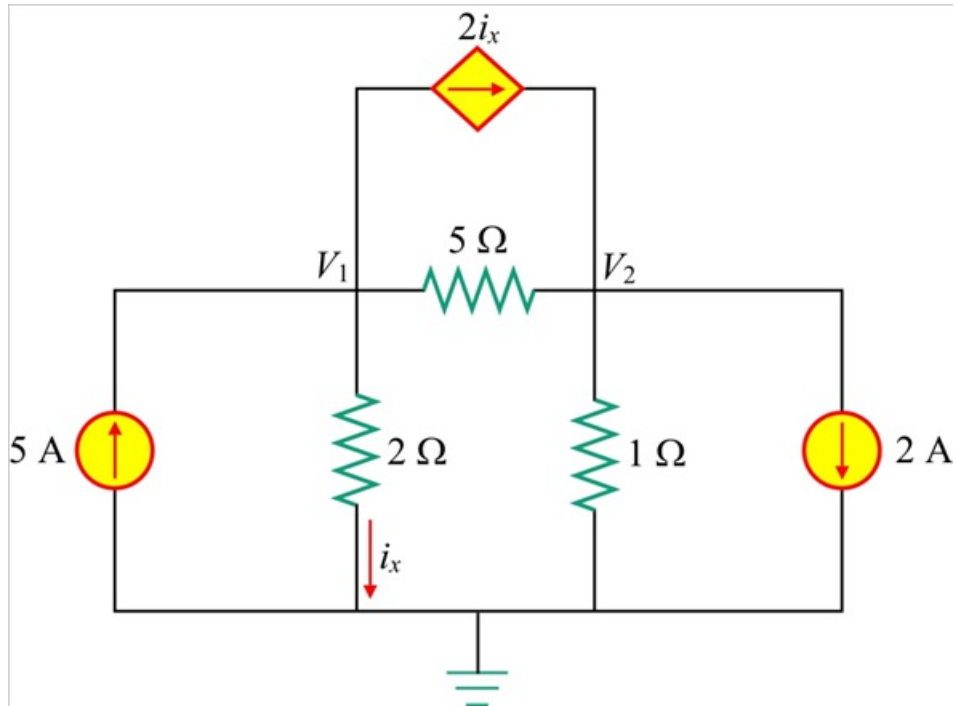


Figure 1

Step 2 of 8

Theoretical analysis:

From the Figure 1,

$$i_x = \frac{V_1}{2} \dots\dots (1)$$

Apply Kirchhoff's current law at node 1.

$$\frac{V_1 - V_2}{5} + \frac{V_1}{2} + 2i_x = 5$$

Substitute $\frac{V_1}{2}$ for i_x in the above equation.

$$\frac{V_1 - V_2}{5} + \frac{V_1}{2} + \frac{2V_1}{2} = 5$$

$$\left(\frac{1}{5} + \frac{1}{2} + 1\right)V_1 - \frac{V_2}{5} = 5$$

$$\left(\frac{2+5+10}{10}\right)V_1 - \frac{2V_2}{10} = 5$$

$$17V_1 - 2V_2 = 50 \dots\dots (2)$$

Step 3 of 8

Apply Kirchhoff's current law at node 2.

$$\frac{V_2 - V_1}{5} + \frac{V_2}{1} - 2i_x = -2$$

Substitute $\frac{V_1}{2}$ for i_x in the above equation.

$$\frac{V_2 - V_1}{5} + \frac{V_2}{1} - \frac{2V_1}{2} = -2$$

$$\left(-\frac{1}{5} - 1\right)V_1 + \left(\frac{1}{5} + 1\right)V_2 = -2$$

$$\left(\frac{-1-5}{5}\right)V_1 + \left(\frac{1+5}{5}\right)V_2 = -2$$

$$-6V_1 + 6V_2 = -10$$

$$3V_1 - 3V_2 = 5$$

$$3V_1 = 5 + 3V_2$$

$$V_1 = \frac{5+3V_2}{3} \dots\dots (3)$$

Step 4 of 8

Substitute $\frac{5+3V_2}{3}$ for V_1 in the equation (2).

$$17\left(\frac{5+3V_2}{3}\right) - 2V_2 = 50$$

$$17(5+3V_2) - 6V_2 = 150$$

$$85 + 51V_2 - 6V_2 = 150$$

$$45V_2 = 150 - 85$$

$$V_2 = \frac{65}{45}$$

$$= 1.44 \text{ V}$$

Step 5 of 8

Substitute 1.44 V for V_2 in the equation (3).

$$\begin{aligned} V_1 &= \frac{5+3(1.44)}{3} \\ &= \frac{5+4.33}{3} \\ &= \frac{9.33}{3} \\ &= 3.11 \text{ V} \end{aligned}$$

Therefore, the values of the node voltages, V_1 and V_2 are 3.11 V and 1.44 V.

PSpice Analysis:

- Open the CIS Demo software.
- Once you are in the capture session frame, click on the menu item File, select new, and then click on Project.
- In the New Project box, type 'selected file name' in the Name text box. Ensure that the Analog or Mixed-signal circuit Wizard is activated.
- Connected the circuit diagram as shown in Figure 1.
- Double click on each component and enter the given values.
- Click the New Simulation Profile icon and enter a name in the Name text box. You will need to enter the appropriate settings for this project in the Simulation
- Setting box. Click the Analysis tab, and select Bias Point from the Analysis type list. Click OK and save the document.
- Click on the run icon. You will observe the A/D Demo screen. When you close this screen, you will observe the bias voltages and currents.

Step 6 of 8

The following is the PSpice circuit diagram:

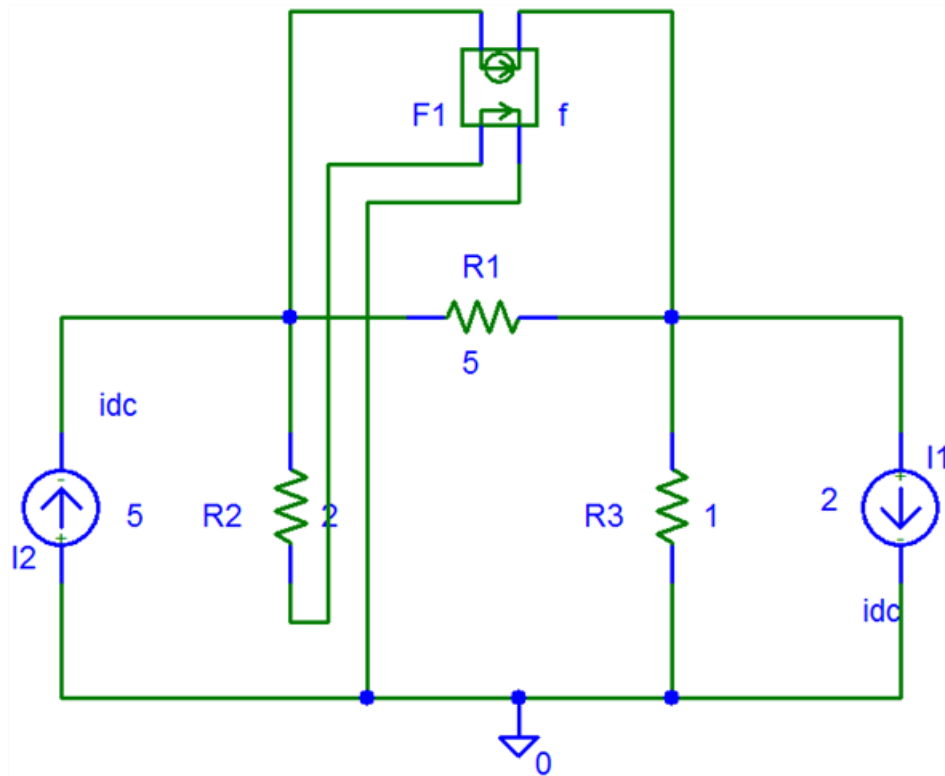


Figure 2

Step 7 of 8

Run the Pspice schematic circuit, the following PSpice circuit with node voltage values.

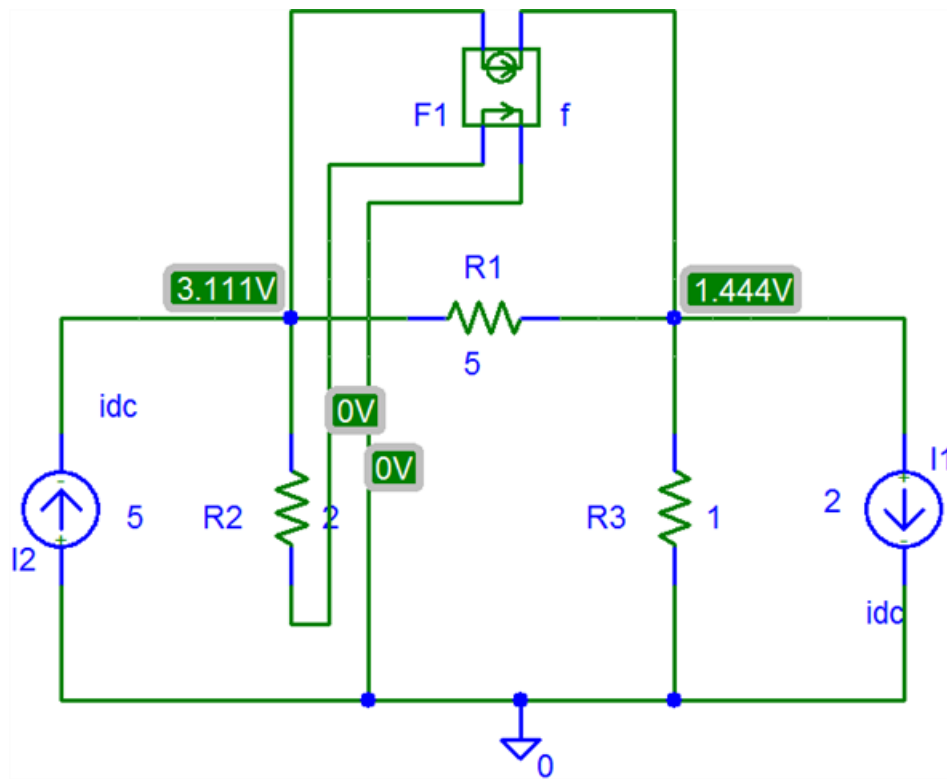


Figure 3

Step 8 of 8

From the Figure 3, the voltage at node 1 is 3.111 V and voltage at node 2 is 1.444 V.

Therefore, the values of the node voltages, V_1 and V_2 are 3.111 V and 1.444 V.

Hence, the node voltages obtained in both theoretical analysis and PSpice are same.